MILITARY SPECIFICATION

TEST SET: TS-3354/PRC-68

This specification is approved for use by the Marine Corps, Department of Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers the requirements for one type radio test set for testing the AN/PRC-68 Radio Set. This test set shall be a self-contained unit designed for portable field use and shall provide quantitative information on the battery condition, receiver sensitivity and transmitter power output. Field strength indication shall also be provided to aid in the alignment of the AN/PRC-68 Radio Set.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-T-152 - Treatment, Moisture - and Fungus-resistant, of Communications, Electronic, and Associated Electrical Equipment

MIL-V-173 - Varnish, Moisture and Fungus Resistant (For Treatment of Communications, Electronic, and Associated Equipment).

MIL-E-16400 - Electronic, Interior Communication and Navigation Equipment, Naval Ship and Shore, General Specification for

MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of

MIL-T-28800 - Test Equipment for use with Electrical and Electronic Equipment, General Specification for

MIL-E-52798 - Enamel, Alkyd, Camouflage

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commandant of the Marine Corps, Headquarters Marine Corps, Code LMA-1, Washington, D.C. 20380 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-108 - Definition of and Basic Requirement for Enclosure for Electric and Electronic Equipment

MIL-STD-129 - Marking for Shipment & Storage

MIL-STD-454 - Standard General Requirements for Electronic Equipment

MIL-STD-810 - Environmental Test Methods

MIL-STD-470 - Maintainability Program Requirements (For Systems and Equipment)

MIL-STD-471 - Maintainability Demonstration

MIL-STD-781 - Reliability Tests Exponential Distribution

MIL-STD-785 - Reliability Program for Systems and Equipment Development and Production

MIL-STD-1275 - Electrical Circuit, 28 VDC, Transient Characteristics for Military Vehicles

MS3116 - Connector, Receptacle, Electric, Series , Solder Type, Jam Nut Mounting, Bayonet Coupling

HANDBOOKS

MILITARY

MIL-HDBK-472 - Maintainability Prediction

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment

... Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

MARINE CORPS
150564 - Front Panel

Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.

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2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precendence.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), prior to the submission of any completed test sets for acceptance, the contractor shall furnish first articles for determination of compliance to this document. Examination and testing shall be as specified herein. Approval of the first article by the procuring activity shall not relieve the contractor of his obligation to supply a product which conforms to this document. Any changes or deviations of production units from the first articles shall be subject to the approval of the contracting officer.

3.2 Parts, materials, and processes. In the selection of parts, materials and processes, the class, grade or type shall be such that, when mounted in the equipment, the part or material will meet the performance requirements specified herein. Fungus-inert materials shall be in accordance with requirement 4 of MIL-STD-454. Where materials which are nutrients for fungi must be used, such materials shall be treated in accordance with MIL-T-152 using varnish conforming to MIL-V-173.

3.3 Design and construction. The test set shall be designed and constructed in accordance with MIL-E-16400 unless otherwise stated herein. Each test set shall consist of the following components:

ITEM	DRAWING PART NUMBER	QUANTITY
Test Set, TS-3354 PRC-68	705559-801	1
Electronic Component Assy	917837-801	1
Test Cable Assy., Antenna	565112-801	1
Test Cable Assy., Audio	466478-801	1
Alignment Tool	805246-1	1
Antenna Assembly	914161-803	1

3.3.1 Test set case. The test set case shall house the test set assembly and shall be provided with a removable hinged cover. The removable hinged cover shall provide storage space for all test set adapters, cable assemblies, connectors and tuning tools. A carrying handle shall be provided on the test set case. The test set shall be mounted within the case using captive hardware. Covers shall incorporate captivated hardware. Seals or sealing surfaces shall be capable of 25 open and close cycles without loss of leakage resistance.

3.3.2 Electrical. The electrical requirements for operating the test set shall be as follows.

3.3.2.1 Input power. The test set shall be powered by a standard AN/PRC-68 battery attached to the battery test connections or by an external 24-volt (V) direct current (dc) power source through a 24-V dc power connector. The test set shall be capable of operating as specified herein from a 24-V dc power source in accordance with MIL-STD-1275. Reverse polarity protection shall be provided.

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3.3.2.2 Input power connector. Two input power connections shall be provided as follows:

(a) Battery input power connector: This connector shall mate with the standard battery connector and shall also serve as the battery test connector.

(b) 24-V dc input connector. This shall be a MS3114E10-6P connector.

3.3.2.3 Overload protection. Overload protection shall be provided by means of a circuit breaker which may be integral to the power switch.

3.3.3 Panel configuration. The panel configuration shall be as shown in Drawing No. 159564.

3.3.3.1 Connectors. The following connectors shall be provided on the front panel:

(a) Battery connector - As specified in 3.3.2.2(a)

(b) Radio set connector - This connector shall mate with the standard AN/PRC-68 Radio Set power connector.

(c) External 24-V dc connector - As specified in 3.3.2.1.

(d) Antenna connector, UG-657 A/U

(e) Audio connector, U-133/U

3.3.3.2 Controls. The following controls shall be provided on the front panel.

(a) Power Switch - 3 positions - OFF-BAT-EXT

(b) Sensitivity Switch - 2 positions - Set and Read (momentary control)

(c) Function Switch - 4 positions - receive (REC), transmit (XMT), battery (BAT), or field strength meter (FSM).

(d) Frequency Select Switches - four separate switches with digital display compatible with AN/PRC-68 Radio Set synthesizer switch readout.

3.3.4 Test set meter. Shall be Army Part No. SM-D-660330 or equal.

3.3.5 Cables. Two radio set and test cable assemblies shall be provided: Cable Assembly, Antenna, Drawing Part No. 565112-801 and Cable Assembly Audio, Drawing Part No. 466478-801.

3.3.6 Antenna. An antenna, Drawing Part No. 814161-803 shall be provided.

3.3.7 Interchangeability. Provisions for interchangeability shall be in accordance with requirement 7 of MIL-STD-454.

3.3.8 Soldering. Soldering shall be in accordance with requirement 5 of MIL-STD-454.

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3.3.9 Burn-in. The test set shall be able to withstand the burn-in test specified in 4.6.7 prior to acceptance for delivery.

3.3.10 Safety. Personnel hazard safety shall be in accordance with Requirement 1 of MIL-STD-454.

3.4 Reliability. The test set shall have a lower test mean-time between failures (MTBF), (θ), of 2000 hours and an upper test MTBF (θ_o) of 4000 hours in accordance with Test IV C of MIL-STD-781.

3.4.1 Reliability program plan. A reliability program plan shall be prepared in accordance with MIL-STD-785. The plan shall include a failure data collection analysis and corrective action system.

3.4.2 Reliability prediction. A reliability prediction shall be preformed in accordance with MIL-STD-785 and section 2 of MIL-HDBK-217. The prediction shall assume a +65° C external ambient air temperature, considering temperature rises within the equipment enclosures and asssume a "Ground, Mobile" environment.

3.4.3 Failure data collection analysis and corrective action. A failure data collection, analysis and corrective action system shall be required. The system shall be described in the reliability program plan.

3.5 Maintainability. The test set shall be constructed to provide ease of maintenance, accessibility and replacement of all modules and parts as primary design considerations. Maintenance and repair shall be accomplished to the maximum extent practicable when the system is operational.

3.5.1 Maintainability program plan. A maintainability program plan shall be prepared in accordance with MIL-STD-470.

3.5.2 Maintainability prediction. A maintainability prediction shall be performed in accordance with Procedure III of MIL-HDBK-472 at each of three levels of maintenance.

3.5.2.1 Organizational level. Corrective maintenance limited to replacement of battery. Maximum time to repair shall be 15 minutes, at the 95th percentile.

3.5.2.2 Intermediate level. Corrective maintenance limited to replacement of battery and modules. Standard test equipment will be used to perform module fault isolation. Maximum time to repair shall be 3 hours, at the 95th percentile.

3.5.2.3 Depot level. Repairs to be made to the piece part level of reparable modules. Normal bench tools and standard test equipment shall be employed. Maximum time to repair shall be 3 hours, at the 95th percentile.

3.6 Performance. The performance requirements of the test set shall be as follows:

3.6.1 Duty cycle. The test shall be capable of continuous operation without degradation of performance.

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3.6.2 Test set functions. The test set shall have four basic functional test capabilities as follows:

(a) Battery Test
(b) Receiver Test
(c) Transmitter Test
(d) Field Strength Indicator

3.6.2.1 Battery test. The battery test circuit shall provide for the measurement of the battery voltage under full load conditions of 330 ± 30 milliamperes at ± 5 Vdc. The function switch shall have spring return from the battery position, and when in this position shall indicate the status of the battery on a scale from 0 to 10. The scale shall be divided so the number 5 is in the center of the scale. Odd numbers shall appear under the scale divisions and shall be colored red to the left of center scale, and green to the right of center. For the battery voltage test, the divisions shall be approximately 1 V per division with two divisions equal to 9 Vdc ±15 percent, 5 equal to 12 Vdc ±5 percent, 10 equal to 17 Vdc ±10 percent. Reverse battery polarity protection shall be provided.

3.6.2.2 Receiver test. In this position, the test set shall provide an indication of the quieting sensitivity of the receiver undergoing tests. A frequency synthesizer shall be included in the test set to allow compatibility with AN/PRC-68 Radio Set frequencies. This will allow sensitivity measurements of any one of the 1000 channels of the radio sets. The test set front panel shall include four frequency selection switches for selection of the desired test frequency. The overall frequency stability of the test set shall be ±0.005 percent over the environmental extreme temperatures specified for the test set. The test set signal generator output shall be attenuated to provide 0.5 +0, -0 microvolt signal into a 50 ohm load at the selected operating frequency. The attenuated signal is coupled through an external cable to the radio set to be used to measure receiver sensitivity. An acceptable quieting sensitivity shall be indicated by the meter reading in the green section while a reading in the red section indicates poor sensitivity of the receiver. The audio meter shall indicate a reduction in audio level necessary to produce mid scale (50 percent) reading when the test set is in the READ position of the receive position. This reduction is in the measure of quieting and shall be 14 ± 1 decibels (dB).

3.6.2.3 Transmitter test. The test set shall include a test circuit which will provide a meter indication of the transmitter radio frequency power output with the antenna coupler bypassed. The test set meter shall be calibrated for a mid-scale indication of 0.5 watt (W) ± 0.1 W and a full scale indication of 1 W ± 0.1 W. The meter test circuit shall be relativity insensitive to frequency over the range of 30 to 75.95 megahertz (MHz) and shall not vary more than ± 0.1 W over this frequency range from a reference of 1 W at 54.05 MHz.

3.6.2.4 Field strength indicator. The test set shall include an untuned test circuit and pick-up antenna which will provide a meter indication of transmitter power output over the frequency range of 30 to 79.95 MHz. A diode limiter shall be included to protect against meter burn out. The sensitivity of this test circuit shall enable the transmitted output of an AN/PRC-68 Radio Set to provide at least 1/2 scale reading when the radio is located from 8 inches to 24 inches from the test set antenna.

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3.6.2.5 Voltage regulator. The test set shall include provisions for being powered by an alternative 24 V dc power source. A voltage regulator circuit shall be included which will maintain a 14 V dc ± 0.5 V dc regulated supply to the test set with an input voltage of 19 to 32 V dc. Short circuit protection of this internal power supply shall be provided.

3.6.3 Environmental requirements. The environmental requirements for the test set shall be as follows:

3.6.3.1 Temperature classification. The temperature classification of the test set shall be Range 1 of Table 1 of MIL-E-16400, except that the operating range shall be -40 C to +55 C.

3.6.3.2 Case. The test set shall incorporate a hinged cover which when secured will make the case watertight in accordance with MIL-STD-108.

3.6.3.3 Vibration. The test set, with cover secured, shall be capable of withstanding the vibration test specified in MIL-STD-810, Method 514.2, Procedure X, Curve AV.

3.6.3.4 Drop. The test set shall be capable of satisfactory operation with no degradation outside of performance limits after completion of tests in 4.6.14. The test set shall be subjected to a series of drops in accordance with MIL-STD-810, Procedure II, Method 516.2 with cover secured. The test set shall withstand this test with no mechanical damage other than superficial case damage.

3.6.3.5 Humidity. The test set, with cover open, shall be capable of passing the humidity test specified in MIL-E-16400.

3.6.3.6 Salt fog. The test set, with cover open, shall be capable of passing the salt fog test specified in MIL-E-16400.

3.6.3.7 Inclination. The test set shall operate normally up to 90 degrees from its level position. The receiver-transmitter and battery shall remain connected to the test set without external support up to 60 degrees from level.

3.7 Dimensions. The maximum dimensions of the test set shall be as follows

Height	7.7 inches
Width	11.5 inches
Depth	8.8 inches

3.8 Weight. The maximum weight of the test set shall be 10 pounds (4.54 kilograms).

3.9 Finish. The test set case shall be finished in accordance with the requirements of MIL-E-16400 except that the finished coat of the case shall be in accordance with MIL-E-52798 (forest green).

3.10 Marking. Marking shall be in accordance with MIL-T-28800.

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3.11 Workmanship. Workmanship shall be in accordance with Requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirement as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows

(a) First article (See 4.3)
(b) Quality conformance (See 4.4)

(1) Production inspection (See 4.4.1)
(2) Production control inspection (See 4.4.2)
(3) Environmental inspection (See 4.4.3)
(4) Inspection of preparation for delivery (See 4.7)

4.3 First article inspection. The first articles shall be subject to all of the examinations and tests indicated in Table I.

TABLE I. Tests.

No.	Test	Req. Para	Test Para	1st Article	Prod Insp	Prod Cont	Environmental
1.	Temperature	3.7.1	4.6.1.1	x			x
2.	Case	3.7.2	4.6.1.2	x			x
3.	Vibration	3.7.3	4.6.1.3	x			x
4.	Drop	3.7.4	4.6.1.4	x			
5.	Humidity	3.7.5	4.6.1.5	x			x
6.	Salt Fog	3.7.6	4.6.1.6	x			
7.	Inclination	3.7.7	4.6.1.7	x			x
8.	Reliability	3.4	4.6.2	x		x	
9.	Maintainability	3.5	4.6.3	x			
10.	Electrical	3.3.2	4.6.4	x			x
11.	Dimensions & Weight	3.8 & 3.9	4.6.5	x			
12.	Interchangeability	3.3.7	4.6.6	x			x
13.	Burn-In	3.3.9	4.6.7	x	x		
14.	Battery Test	3.6.2.1	4.6.8.1	x	x		
15.	Receiver Test	3.6.2.2	4.6.8.2	x	x		
16.	Transmitter Test	3.6.2.3	4.6.8.3	x	x		
17.	Field Strength Indicator	3.6.2.4	4.6.8.4	x	x		
18.	Voltage Regulator	3.6.2.5	4.6.8.5	x	x		
19.	Mechanical Inspection	4.5	4.5	x	x		
20.	Inspection of Preparation	5	4.7	x	x		

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4.4 Quality conformance inspection. The quality conformance inspection shall consist of the following.

4.4.1 Production inspection. Production inspection shall be made on every test set offered for delivery. The inspection shall consist of examination and testing which will prove the workmanship and reveal the omissions and errors of the production process such as functional and performance tests from design, tests of adjustment, and tests which detect hidden defects of material. Production inspection shall include the tests indicated in Table I.

4.4.2 Production control inspection. Production control inspection, including sampling, shall consist of tests indicated in Table I and the inspection procedures of MIL-STD-105 using the special inspection levels. The inspection level shall be S-3 for normal, tightened, and reduced inspection. The acceptable quality level (AQL) shall be 6.5 percent. A lot is defined as one month's production of test sets. Production control inspection shall be performed on test sets which have passed production inspection. The test sets shall satisfactorily meet the requirements of production control inspection prior to release for shipment.

4.4.2.1 Rejected lots. If an inspection lot is rejected, the supplier may withdraw the lot from further inspection. The supplier may also rework a rejected lot to correct the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.3 Environmental inspection. Environmental inspection shall consist of tests indicated in Table I and shall be performed on test sets which have been subjected to and have passed production inspection.

4.4.3.1 Sampling for environmental inspection. Three test sets shall be selected from each successive 100 test sets produced or fraction thereof. The first sample test sets shall be selected from the first month's production. Environmental tests shall be distributed within each sample of three test sets as follows:

EQUIPMENT	TEST NUMBER
Test Set No. 1	1, 10
Test Set No. 2	2, 5, 12
Test Set No. 3	3, 7

4.4.3.2 Nonconforming environmental sample test sets. If a sample test set fails the inspection specified in 4.4.3, the contractor shall immediately investigate the cause of failure and shall report to the quality assurance representative the results thereof and details of the corrective action taken to correct units of product which were manufactured under the same conditions with the same materials, processes and so forth. If the quality assurance representative does not consider that the corrective action will enable the product to meet requirements specified herein, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer.

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4.4.3.3 Reinspection of conforming environmental sample units. Sample units which have been subjected to, and passed, environmental tests may be accepted on the contract provided they are re-inspected to, and pass, production inspection after repair of all damage.

4.5 Inspection. Inspections shall be performed to determine conformance to the requirements specified herein. The test set shall be given a thorough mechanical and visual inspection to determine that the quality of all materials and workmanship is in compliance with the requirements specified herein and in MIL-E-16400. Particular attention shall be given to the following:

(a) Completeness
(b) Nameplates, identification markings and labels.
(c) Ease of operation of all front panel controls, switches and mating connectors.
(d) Finishes
(e) Welded joints, if any
(f) Solder joints
(g) The fit of components in their respective positions
(h) Cover fasteners
(i) Fit and condition of seals
(j) Other visual defects

4.6 Tests

4.6.1 Environmental. Where reference measurements are required for a particular environmental test, the following performance tests shall be performed.

Test No. 14	Battery Test
Test No. 15	Receiver Test
Test No. 16	Transmitter Test
Test No. 17	Field Strength Indicator
Test No. 18	Voltage Regulator

4.6.1.1 Temperature. The test set shall be subjected to and pass the temperature test specified in MIL-E-16400, except that the temperature extremes shall be as specified herein.

4.6.1.2 Case. The test set with covers installed and secured, shall be subjected to and pass the watertightness test specified in MIL-STD-108.

4.6.1.3 Vibration. The test set shall be subjected to and pass the test specified in MIL-STD-810, Method 514.2, Procedure X, Curve AV.

4.6.1.4 Drop. The test set shall be subjected to and pass the test specified in MIL-STD-810, Method 516.2.

4.6.1.5 Humidity. The test set shall be subjected to and pass the test specified in MIL-E-16400.

4.6.1.6 Salt fog. The test shall be subjected to and pass the test specified in MIL-E-16400.

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4.6.1.7 Inclination. Operation of the test set up to the limits specified in 3.7.7 shall be demonstrated.

4.6.2 Reliability

4.6.2.1 Reliability test plan document. The contractor shall prepare a reliability test plan document in accordance with MIL-STD-781.

4.6.2.2 Reliability qualification. The contractor shall conduct a reliability qualification test on the first articles. The test shall be conducted in accordance with Test Plan IV C of MIL-STD-781 except that the temperature range shall be as specified herein.

4.6.2.3 Reliability sampling. Eight test sets shall be randomly selected, by the Government Inspector, from every lot of 100 test sets produced, or fraction thereof for testing in accordance with Test Plan IV C of MIL-STD-781, except that the temperature range shall be as specified herein.

4.6.2.4 Modes of operation and test parameters. The test set shall operate continuously from an external 24 V dc source with the function switch in the receiver position. Once every 24 hours, compliant operation of the battery test, receiver test, transmitter test and field strength indication functions shall be verified. MIL-STD-781 is applicable.

4.6.2.5 Failure. Failure is defined to occur any time the test set fails to meet any of the requirements specified herein associated with the above verification measurements.

(a) Nonrelevant failures - those failures which (1) are test equipment orientated, (2) are externally caused or (3) are censored by the procuring activity as a result of Government approved corrective action. These features shall not be countable during reliability tests.
(b) Relevant failure - All other failures are defined as relevant.
(c) MIL-STD-781 shall apply.

4.6.2.6 Reliability test procedures. The detailed test conditions, procedures and methods for the tests described in 4.6.2.2 and 4.6.2.3 above shall be in accordance with the approved reliability test procedure for the TS-3354/PRC-68. MIL-STD-781 shall apply. No single equipment "ON" time shall be less than one-half the average operating time of all equipments on RST. Operation shall be representative of the operations expected in the field. Occurrence of pattern failures shall be cause for rejecting the test lot.

4.6.3 Maintainability. A maintainability demonstration plan shall be prepared by the contractor in accordance with MIL-STD-471, to satisfy requirement of a 15 minute maximum time to repair, as specified in 3.5.

4.6.4 Electrical. Compliance with the requirements of 3.3.2 shall be demonstrated.

4.6.5 Dimensions and weight. Compliance with the requirements of 3.8 and 3.9 shall be demonstrated.

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4.6.6 Interchangeability. Compliance with the requirements of 3.3.7 shall be verified. The interchangeability shall be verified by substitution of like plug-in modules or subassemblies.

4.6.7 Burn-in. Prior to production acceptance tests, all test sets shall be operated for six 8-hour temperature cycles, each cycle consisting of 4 hours at -40°C and 4 hours at +65°C. During the last 20 minutes of each hour of the high temperature cycle, the test set shall be vibrated at 2 g's at a frequency of 30 hertz. The rate of temperature change shall not be less than 5°C per minute.

4.6.8 Performance.

4.6.8.1 Battery. Compliance with the requirements of 3.6.2.1 shall be demonstrated.

4.6.8.2 Receiver. Compliance with the requirements of 3.6.2.2 shall be demonstrated on six frequencies 30.05 MHz, 43.05 MHz, 53.95 MHz, 54.05 MHz, 67.05 MHz, and 79.95 MHz.

4.6.8.3 Transmitter. Compliance with the requirements of 3.6.2.3 shall be demonstrated at the same frequencies specified in 4.6.8.2.

4.6.8.4 Field strength indicator. Compliance with the requirements of 3.6.2.4 shall be demonstrated for the same frequencies specified in 4.6.8.2.

4.6.8.5 Voltage regulator. Compliance with the requirements of 3.6.2.5 shall be demonstrated.

4.6.9 Safety. The contractor shall document design actions introduced to facilitate compliance with requirements of 3.3.10.

4.7 Inspection of preparation for delivery. Inspections shall be conducted to ensure conformance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Perservation, packaging and packing shall be the level of shipment specified in the contract or order in accordance with MIL-E-17555 (see 6.2).

5.2 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The TS-3354/PRC-68 Test Set is used to provide the following functional tests of the AN/PRC-68 Radio Set:

(a) Battery Test
(b) Receiver Test
(c) Transmitter Test
(d) Field Strength Indicator

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The field strength indicator provides a means to tune the radio to any desired frequency.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specfication.
(b) When first article is not required (see 3.1).
(c) Levels of preservation, packaging and packing required (see 5.1).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n)(2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph No.	Data requirements	Applicable DID no.	Options
3.1	First Article Inspection Procedures	DI-T-4901	
3.1	First Article Inspection Report	DI-T-4902	
3.4.1	Reliability Program Plan	DI-R-7079	
3.4.2	Report, Reliability Prediction	DI-R-7080	
3.4.3	Failure Analysis and Corrective Action Report	DI-R-22930	
3.5.1	Maintainability Program	DI-R-2127	
3.5.2	Maintainability Analysis and Prediction	DI-R-2128	
4.4.1	Production Inspection Procedures	DI-T-4904	

(Copies of DID's required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. The first article shall be examined and tested for approval at the contractors's plant or at an independent commercial testing laboratory acceptable to the procuring activity. First article tests shall be witnessed by Government representative of the procuring activity.

Preparing activity:
Navy - MC
Project No. 5625-N802

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MIL-T-29402(MC) TEST SET: TS-3354/PRC-68

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